How accurate LIGHT v0_1 is

The basic network of LIGHT v0_1, version 0.1.1, on 29 September 2026, with its energy ladder, its system map and its reaction rates

Bottom line. LIGHT v0_1 reproduces the fitted KrF decays [1, 2] within 2.3 to 6.1 % rms and predicts decays that no fit used [4] within 11.7 % rms, and at the Tahoma laser [5] its default network, with n_e free, gives 93 % of the measured small-signal gain of 6.39 %/cm.

With the optional Kr_2^* group on as well, as ML scores it, v0_1 gives 95 % of the gain, 110 % of the saturation intensity and 106 % of the formation efficiency, and without the group's estimated Kr_2^* + F row 89 % of the gain. The C/B emission ratio of Yu and Setser [3] is matched to -0.5 %. No fit used Tahoma's numbers and no rate was set by a laser measurement, so the Tahoma scores are predictions.

Its largest misses are on data that conflict with the fitted sets, 19.3 and 34.1 % rms on the C-A decays of Hakuta et al. [6] and -22.3 % on the KrF(B) lifetime of Burnham and Searles [7], and the VUV yield of pure krypton at low pump, 136 % of Morozov's measurement [8].

[1] Eden, Waynant, Searles and Burnham, Appl. Phys. Lett. 32, 733 (1978). [2] Gadomski, Xu, Setser and Nelson, Chem. Phys. Lett. 189, 153 (1992). [3] Yu and Setser, J. Phys. Chem. 94, 2934 (1990). [4] Xu, Gadomski and Setser, J. Chem. Phys. 99, 2591 (1993). [5] Spectra Technology, *Experimental Support for KrF Laser Modeling*, final report to Los Alamos, 19 July 1989, OSTI 10136715. [6] Hakuta et al., Jpn. J. Appl. Phys. 29, 1194 (1990). [7] Burnham and Searles, J. Chem. Phys. 67, 5967 (1977). [8] Morozov, Heindl, Krücken, Ulrich and Wieser, J. Appl. Phys. 103, 103301 (2008).

LIGHT v0_1 is v0's working copy. It takes one input, a power deposited uniformly in the electrons, and its basic network carries the charge chemistry and fluorine recombination, 37 reaction rows in all. The KrF(B,C) block is v0's. Every KrF rate in it but one is a measurement of that process or a joint fit to kinetic measurements of it, and the exception is the KrF(B) radiative lifetime, Kannari's 6.99 ns where Eden measured 6.8 $\pm$ 0.2 ns. The rows v0_1 changed carry measured values where a measurement exists. No rate was set by a laser measurement. Scored against Tahoma, the one independent Kr/F_2 laser, under the same beam drive as v0, v0_1 gives the small-signal gain at $\times 0.95$ the measured value, the saturation intensity at $\times 1.10$ and the formation efficiency $g_0 I_{sat}/P_{in}$ at $\times 1.06$. That score includes the optional Kr_2^* group, and without its estimated Kr_2^* + F row the gain is $\times 0.89$. The default network, without the group, gives $\times 0.93$. The largest open questions are the trimer rate above 425 K and the rate of Kr_2^* + F $\rightarrow$ KrF(B) + Kr, and neither has been measured. Table 1 lists every test. The comparison with v0 is in `docs/REVIEW/`.

1 How the numbers were made

Each dataset has one role. Kinetic measurements of the channels being fitted train them. Other kinetic measurements are held out and test the fit. Laser performance only scores, and the fitting code refuses a dataset of that kind. This is the repository's predictive-only policy. Measurements score the model after the fact and never steer a number. Every number in this note is written by `docs/make_overview_v0_1.py` and `docs/make_system_map_v0_1.py` from v0_1 and from `ML/`, and a test fails when those files no longer match the model.

ML runs v0_1 when the environment variable `LIGHT_VERSION` names it and v0 otherwise, and v0's own overview tests show that the switch leaves every v0 number unchanged. v0_1 takes only a power input. Tahoma was pumped by an electron beam, so v0_1 is scored there under the same prototype beam drive as v0, a test fixture in `docs/` and not part of the model. Section 6 runs v0_1's own input at Tahoma's conditions.

test	measured	v0_1	role of the data
energy ledger closure, 9 runs across the modes		below 10^{-12} of E_{dep}	bookkeeping
KrF(B) decay against Kr and F_2, Eden 1978		rms 2.3 and 3.2 %	train
KrF(B,C) slow decay, Gadomski 1992 Figs. 1 and 2		rms 6.1 and 4.2 %	train
C/B emission ratio at 300 K, Yu and Setser 1990	0.16 ± 0.02	-0.5 %	train
KrF(B,C) decays, Xu 1993 Figs. 5 to 7		rms 11.7 %	held out
decay constants Xu 1993 printed, 50 and 100 Torr Kr	5.0×10^{7} and 6.2×10^{7} s^{-1}	$\times0.91$ and $\times0.97$	structure check
C-A decays, Hakuta 1990		rms 19.3 and 34.1 %	held out, in conflict
τ_B, Burnham and Searles 1977	9.0 ± 0.6 ns	-22.3 %	held out, in conflict
C/B ratio 230 K over 300 K, Yu and Setser	1.30 ± 0.06	-8.4 %	held out, partly behind the gap v0_1 books
Kr^* + Kr^+ made per eV, pure Kr and Ar, Saito 2003		$\times1.02$ and $\times1.03$	check
Kr_2^+ + e^- recombination, Shiu and Biondi 1977		$\times1.00$	v0_1's input
$Kr^+ + 2Kr \rightarrow Kr_2^+ + Kr$, Smith et al. 1972	2.74×10^{-31} to 3.3×10^{-31}	2.4×10^{-31}	in v0_1's input
VUV yield of pure Kr at low pump, Morozov 2008	0.42 ± 0.05	0.57, $\times1.36$	check
electron closure at Tahoma, kinetic EEDF loop	T_e 1.61 eV	1.60 eV	reference
Tahoma small-signal gain g_0	6.39 %/cm	6.10, $\times0.95$	device, score only
the same, default network without Kr_2^*		5.94, $\times0.93$	device, score only
the same, trimer on $T_g^{-1/2}$		3.02, $\times0.47$	device, score only
the same, without Kr_2^* + F $\rightarrow$ KrF		5.69, $\times0.89$	device, score only
Tahoma saturation intensity I_{sat}	1.4 ± 0.1 MW/cm^2	1.54, $\times1.10$	device, score only
Tahoma formation efficiency $g_0 I_{sat}/P_{in}$	0.27	0.285, $\times1.06$	device, score only

Table 1: LIGHT v0_1 against the data. Rates are in cm^{3}s^{-1} or cm^{6}s^{-1}. A percentage is v0_1's deviation from the data, signed for a single value and rms for a set. A $\times$ factor is v0_1 over the measurement.

2 The KrF(B,C) block

v0_1 keeps v0's KrF(B,C) block. KrF has two emitting states. B lases at 248 nm, and C radiates slowly in a band at 275 nm. LIGHT treats them as the two coupled levels of eq. 5 of Gadomski et al. (1992) and reproduces that model's analytic decay rate to within 1.6×10^{-4} at every pressure tested. The block has 10 constants. Nine come from one joint fit to three measurement sets that agree with one another, the KrF(B) decays of Eden et al. (1978), the Kansas State B/C decays (Gadomski et al. 1992) and Yu and Setser's equilibrium C/B ratio (1990). The tenth, τ_B, stays at Kannari's 6.99 ns, inside the fit's interval of 6.79 to 7.22 ns. v0_1 shares it between the 248-nm B-X band and B-A emission at 272 nm, which by Dunning and Hay's calculation carries 3.5 % of the B-X rate, so the total radiative rate of B is unchanged. The fit's median and 68 % interval for B-C mixing by Kr is $k_{BC} = 9.8\times10^{-12}$ [4.63×10^{-12}, 1.73×10^{-11}] cm^{3}s^{-1}. The equilibrium ratio is $K_{eq} = 1.07$ [0.84, 1.38], and the C lifetime is $\tau_C = 50$ [41, 63] ns. LIGHT carries the best fit, $k_{BC} = 1.05\times10^{-11}$ cm^{3}s^{-1}, $K_{eq} = 1.10$ and $\tau_C = 50$ ns. At 1 atm and 300 K the lasing B state holds 0.69 [0.57, 0.80] of the manifold, and LIGHT's rounded constants give 0.68.

The fitted decay sets are matched to between 2.3 and 6.1 % rms. The held-out decays of Xu et al. (1993) are matched to 11.7 %, and the two decay constants they print come out at $\times0.91$ and $\times0.97$. Hakuta et al. (1990) measured faster C-A decays, missed by 19.3 and 34.1 % rms, and their data conflict with the three fitted sets. Burnham and Searles (1977) measured $\tau_B = 9.0$ ns. Those two conflicts are v0's and remain.

The energy ledger now books Yu and Setser's measured B-C gap, 120 cm^{-1}, where v0 kept REBUILD's 80. It sets how K_{eq} changes with temperature, and their C/B ratio between 230 and 300 K now comes

out $-8.4\,\%$. That is no longer an independent test. Their $120 \pm 50\,\mathrm{cm^{-1}}$ is a consensus of two routes, 110 ± 20 from the absolute ratio and 180 ± 30 from this very pair of temperatures.

Tahoma's gain is the sharpest test of the block, since g_0 is the KrF(B) density times the stimulated cross section. Over 40 draws from the fit's posterior the gain is $5.79\,\%/\mathrm{cm}$ [5.38, 6.63] against the measured 6.39.

3 Charge, recombination and fluorine

The charge chemistry is part of v0_1's basic network, and no electrons are injected. n_e starts from a seed of $10^{15}\,\mathrm{cm^{-3}}$ and follows ionization, attachment and recombination. At the reference point it runs at $1.95\times10^{14}\,\mathrm{cm^{-3}}$ while F_2 lasts and peaks at $3.66\times10^{15}\,\mathrm{cm^{-3}}$ after F_2 burns out at 613 ns. Kr^+ turns into Kr_2^+ at $2.4\times10^{-31}\,\mathrm{cm^6 s^{-1}}$, the mean of three measurements, and Kr_2^+ recombines with electrons at Shiu and Biondi's measured $1.6\times10^{-6}\,(300\,\mathrm{K}/T_e)^{0.55}\,\mathrm{cm^3 s^{-1}}$. Each recombination returns a Kr^* and heats the gas. Of the ions that recombine, 95.0 % meet an electron and 5.0 % meet F^- and form KrF(B), which makes 5.6 % of the KrF formed. Once the F_2 is gone, Penning ionization of Kr^* and recombination hand the same energy back and forth, so the flows through Kr^* and the ions pass 100 % of E_{dep} and most of that energy ends as heat. The gas ends at 762 K.

F + F + Kr gives F atoms back to F_2 at $2.8\times10^{-34}\,\mathrm{cm^6 s^{-1}}$, Cohen and Bott's evaluated Ar value, since no Kr measurement exists. At the reference point it returns 0.1 % of E_{dep} from the F atoms to heat, far too slowly to refill the F_2.

4 KrF formation

The chain from Kr^+ and Kr^* to KrF is v0's. Kr^* + F_2 reacts at $8.47\times10^{-10}\,\mathrm{cm^3 s^{-1}}$, the mean of three first-hand measurements. Velazco, Kolts and Setser (1976) measured the KrF^* yield of that reaction as 90 %, and Yu and Setser (1990) put 38 % of that KrF^* in C. Tahoma measured $g_0 I_{\mathrm{sat}}/P_{\mathrm{in}} = 0.27$. ML puts a 248-nm probe flux on v0_1 and fits its I_{sat} at fixed g_0, as Tahoma did. That gives 0.285. Its KrF formation rate in 5-eV quanta is 0.318 of the pump. The difference is KrF that forms in C, or mixes into it, and decays there without meeting the laser field.

Table 2 moves one link at a time to another value it could take. The measured range of the harpoon yield, 0.70 to 1.0, moves $g_0 I_{\mathrm{sat}}/P_{\mathrm{in}}$ from 0.256 to 0.299 and brackets Tahoma's 0.27. Three links have no measurement at all. They are the KrF^* yield of ion-ion recombination with F^-, the share of that KrF formed in C, and Kr_2^* + F $\rightarrow$ KrF. Only analogs exist for the first two, and both analogs lower $g_0 I_{\mathrm{sat}}/P_{\mathrm{in}}$ toward the measurement. Removing the third lowers it too. The predictive-only policy lets no analog set a link and no link be chosen by its fit to Tahoma. The rest of the Kr_2^* route moves $g_0 I_{\mathrm{sat}}/P_{\mathrm{in}}$ by 0.006 at most.

link	v0_1	tried	what the value is	$g_0 I_{sat}/P_{in}$	change
KrF* yield of $Kr^* + F_2$	0.90	0.70	the measured 0.90 less its stated 0.20	0.256	−0.029
		1.0	the published assignment	0.299	+0.014
		0.5	as for $Ar^* + F_2$	0.227	−0.058
share of that KrF formed in C	0.38	0	if nascent C relaxes to B at 1 atm	0.307	+0.022
KrF* yield of ion-ion recombination	1	0.64	as for Cl^-	0.242	−0.043
share of that KrF formed in C	0	0.38	as for SF_6^-	0.268	−0.017
ion-ion recombination rate	Flannery	×0.38	Avco's model input	0.283	−0.002
$Kr_2^* + F \rightarrow KrF(B) + Kr$	3.0×10^{-10}	0	no measurement exists	0.268	−0.017

Table 2: Each formation link moved alone from v0_1's chain at Tahoma, 2.8 Torr of F_2 in Kr at 665 Torr in all, pumped at 330 kW cm^{-3}. v0_1's chain gives 0.285, and Tahoma measured 0.27.

5 The electron side

v0_1 integrates each krypton electron-impact rate over v0's EEDF with two temperatures, a Maxwellian bulk at T_e and a tail above 9.915 eV read from kinetic EEDFs. Each captured electron takes the energy it really carries. For F_2 attachment at Tahoma that is 0.55 eV, where $\frac{3}{2}T_e$ would book 2.41 eV. Kr_2^+ recombination goes as $T_e^{-0.55}$, so a recombining electron takes 0.95 T_e. A kinetic loop that iterates v0_1 with a Boltzmann solution, with the same recombination law as its sink, is the reference. At Tahoma v0_1's own closure lands within 0.7 % of the loop's T_e and 2.5 % of its n_e, and within 0.9 and 0.5 % of its gain and formation.

The cross sections are v0's. With Biagi's krypton set the computed W is 0.96 to 1.03 times Nakamura's measured values. For F_2 attachment the groups that measured the rate constant disagree beyond their errors, and SIGLO gives between 0.48 and 2.41 times their values.

On pure gases the beam drive's Spencer-Fano yields make 1.02 times the Kr* and Kr^+ per eV that the measured W and N_{ex}/N_i give in Kr, and 1.03 times in Ar. v0_1 takes its Kr_2^+ recombination from Shiu and Biondi and its Kr^+ conversion from a mean that includes Smith et al., so two of v0's pure-gas checks are now v0_1's inputs. The VUV yield of pure Kr at low pump is still the one miss. v0_1 gives 0.57 where Morozov et al. (2008) measured 0.42 ± 0.05. Without the thermal electrons' excitation it gives 0.55, close to the one photon per excitation and per ion that the measured W and N_{ex}/N_i allow, 0.54, so the shortfall is either physics LIGHT lacks or a systematic error of the measurement.

6 The power input

v0_1 feeds its power to the electron gas. Run that way at Tahoma's conditions, with no beam, it gives g_0 = 6.56 %/cm, ×1.03 the measured value. Its electrons then carry the power at T_e = 3.8 eV and $n_e = 2.55\times10^{14}$ cm^{-3}, where the beam-sustained plasma sits at 1.6 eV and 6.3×10^{14} cm^{-3}. The gain agrees, but the electron gas does not resemble the beam's, so the agreement does not test the power input. No discharge-pumped Kr/F_2 measurement has been scored.

7 The energy ladder

Figure 1 places every species of v0_1 at its formation energy, the energy above ground-state Kr and F_2 at rest that Appendix A lists, and draws the rows that move energy between them. It follows the ladder view of the website's excimer simulation, with v0_1's own numbers. KrF(B) lies 5.54 eV below

the ion pair $Kr^+ + F^-$, the measured $D_e(B)$ that replaces REBUILD's estimate, and 5.06 eV above Kr + F. The 248-nm photon carries 5.00 eV of that and leaves KrF(X) only 0.06 eV above Kr + F. The harpoon G3 heats the gas by 3.26 eV for each KrF(B) it makes, the ion-ion recombination K26 by 5.54 eV and the trimer's formation by 0.58 eV. Kr_2F^* emits 2.95 eV at 420 nm and leaves 1.53 eV of lower-state repulsion. The Kr_2^* route is an optional group in v0_1, drawn dashed.

8 The system map

The map on the next page draws v0_1's basic network at the notes' reference point, 1 atm of Kr with 0.5 % F_2, 300 kW cm^{-3} for 1 μs, with n_e seeded at 10^{15} cm^{-3} and then free. Band widths are proportional to the energy they carry, and every box conserves it. Of the deposited energy, 24.8 % passes through KrF in 5-eV quanta. Of that KrF, 19.1 % radiates at 248 nm, 61.2 % forms the trimer and 6.8 % is quenched by electrons, so 4.7 % of the deposited energy leaves at 248 nm.

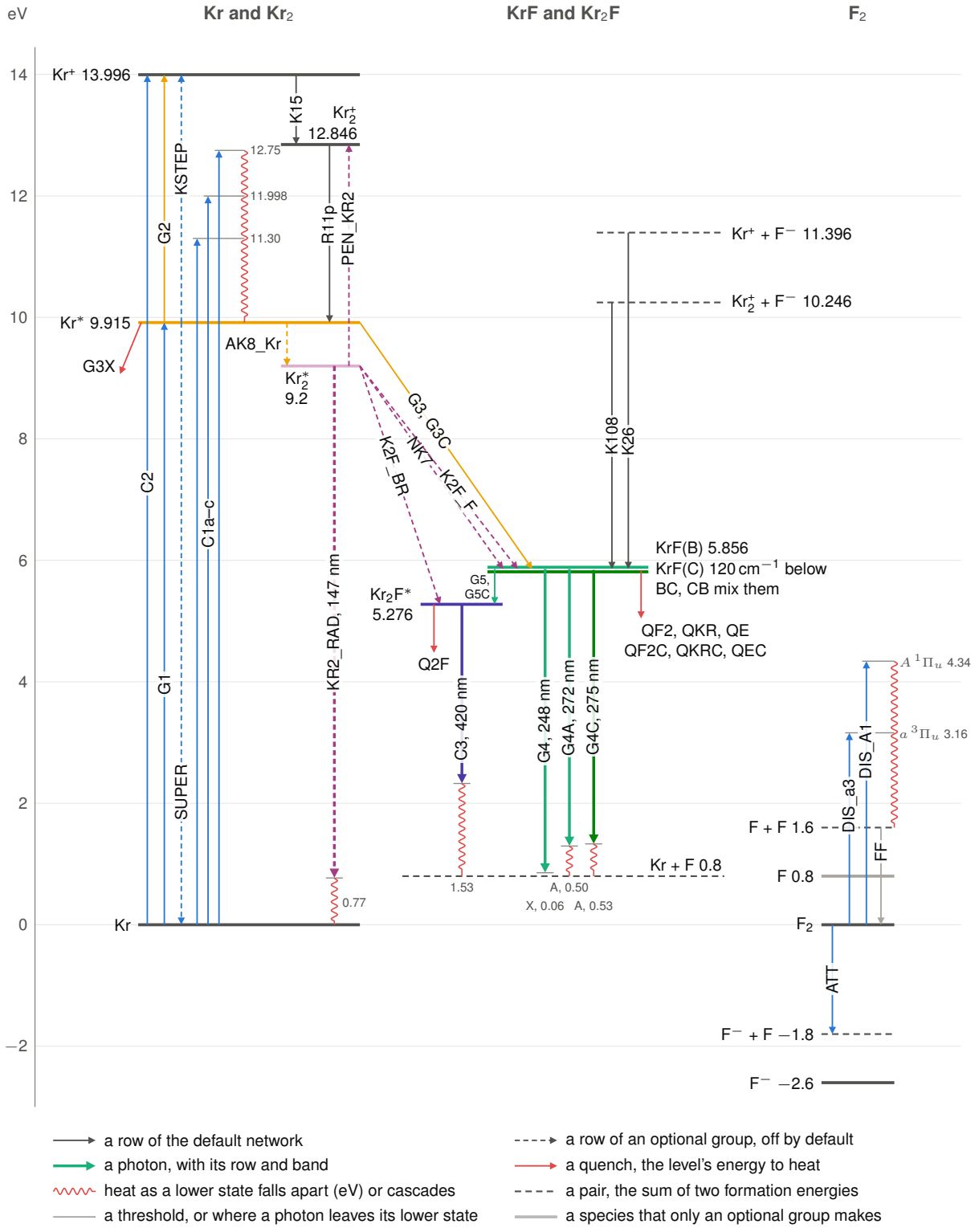
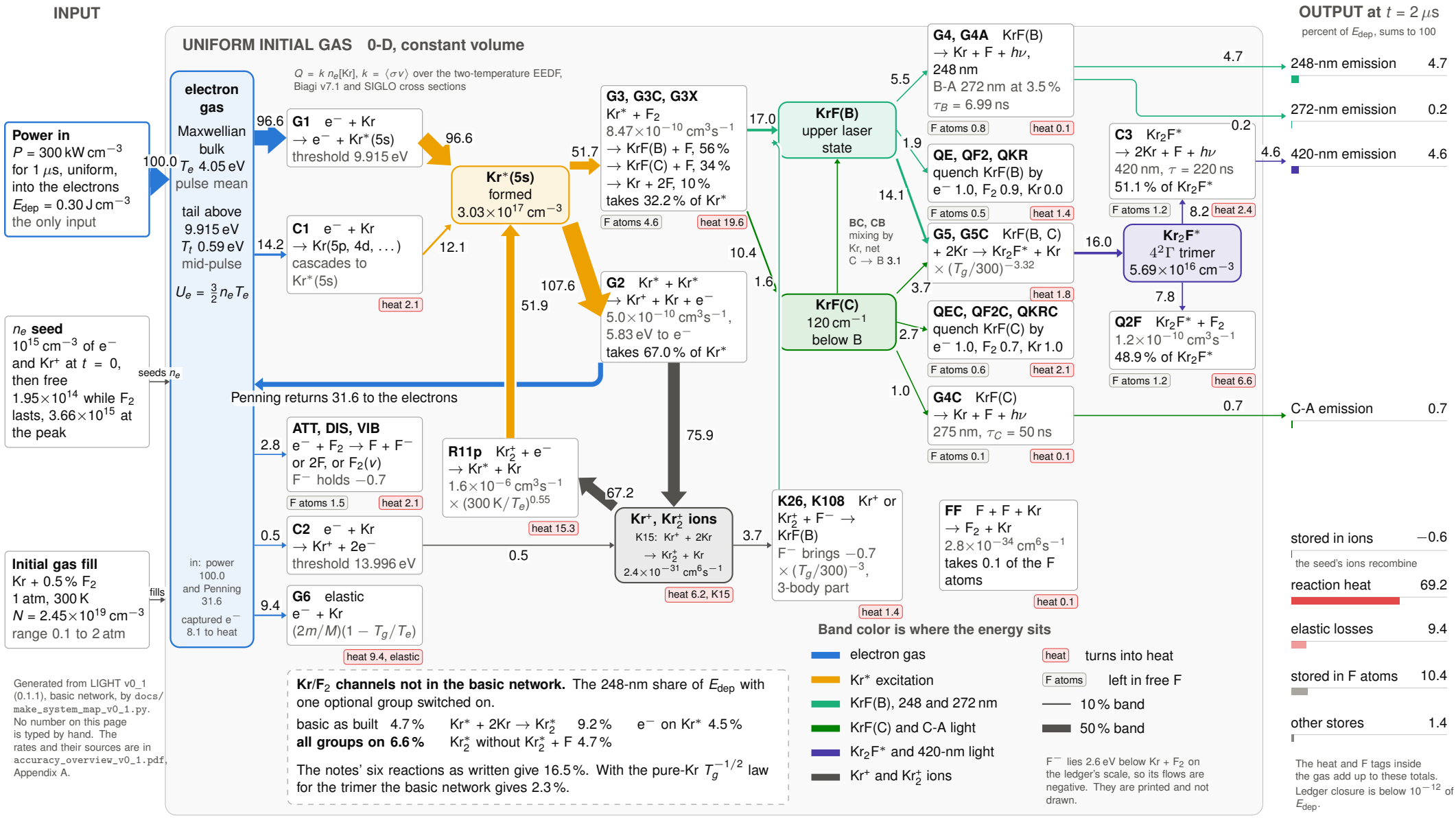


Figure 1: The energy ladder of LIGHT v0_1, version 0.1.1, in eV above ground-state Kr and F_2 at rest. A bar is a species at its formation energy and a dashed bar a pair, the sum of two. An arrow is a row, labelled with its id and solid in the default network. The rows of the optional groups `kr2` and `electron` are dashed, and Kr_2^*, which only they make, is pale. A photon's arrow drops by its quantum, and the red squiggle below it is the heat its lower state gives the gas as it falls apart. A red stub is a quench. The energy a row releases heats the gas, except a photon's, which leaves, and that of G2, PEN_KR2 and SUPER, which goes to the electrons. KrF(B) and KrF(C) lie 120 cm^{-1} apart and are drawn as a pair of bars that every B and C row meets. The rows G6, VIB1, VIB2, VIB3 and VIB4 move nothing between levels and are not drawn.

LIGHT system map, v0_1: energy flow through the Kr/F_2 kinetics

Reference point: 1 atm of Kr with 0.5 % F_2 at 300 K, P = 300 kW cm^{-3} for 1 μs so that E_{dep} = 0.30 J cm^{-3}, n_e seeded at 10^{15} cm^{-3} and then free, run to 2 μs. Band width is proportional to the energy flow. Numbers are percent of E_{dep}, rounded to 0.1.



The notes' outputs as flows, which overlap: to Kr* 110.8 %, to Kr$^+$ 76.5 %, to KrF(B) and KrF(C) 24.8 %, heat 69.2 %, elastic losses 9.4 %, emission 10.2 %. They pass 100 % because recombination hands the ions' energy back to Kr*. Of the ions that recombine, 95.0 % meet an electron and return to Kr*, and 5.0 % meet F^- and form KrF(B). Of the KrF formed, 9.30×10^{16} cm^{-3}, 5.6 % came from ion-ion recombination. Of it 19.1 % radiates at 248 nm, 0.7 % at 272 nm and 3.3 % in the C-A band, 61.2 % forms the trimer, and electrons, F_2 and Kr quench 6.8 %, 5.6 % and 3.4 %. At the end of the run T_g = 762 K and p = 2.55 atm, and F_2 fell below 10 % of the fill at 613 ns.

9 Where v0_1 is not yet reliable

- The trimer above 425 K. KrF + 2Kr $\rightarrow$ Kr_2F^* takes 61.2 % of the KrF at the reference point, where the gas ends at 762 K. The $(T_g/300)^{-3.32}$ law is Shui's calculation, measured only up to 425 K. With the pure-Kr $T_g^{-0.5}$ law, the weakest the data allow, the reference point sends 2.3 % out at 248 nm instead of 4.7 %. Tahoma's window sits at 448 K, and there that law gives the gain at $\times 0.47$ the measured value. Tahoma's gain counts against the weak law as a score and sets nothing.
- Hot gas. Recombination returns the ionization energy as heat. At 0.1 atm and 500 kW cm^{-3} the gas reaches 3461 K, beyond every rate law's data, and Kr_2F^* and Kr_2^+ redissociation are not in the model. The report warns above 1000 K.
- Ion-ion recombination with F^-. Its rate was measured only below 300 Torr, where Jacob et al. (1978) fitted KrF* pulse shapes and v0_1's form, screened as there, lies somewhat above theirs. Rokni et al. (1977) found that $Kr_2^+ + F^-$ forms KrF* as efficiently as $Kr^+ + F^-$, but from relative intensities. The absolute KrF* yield, its B/C split and the rate near 1 atm have no measurement. Table 2 shows what the analogs would do.
- $Kr_2^* + F \rightarrow KrF(B) + Kr$. The optional Kr_2^* group has six rows. Four rest on measurements, although the even split of $Kr_2^* + F_2$ between its two products is Kannari's. $Kr_2^* + F$ is Kannari's estimate, 3.0×10^{-10} cm^{3}s^{-1}, and $Kr_2^* + Kr_2^*$ comes from an unpublished report. At the reference point the group raises the 248-nm share from 4.73 to 9.17 %, and without $Kr_2^* + F$ it gives 4.65 %, so the whole gain rests on the estimate. 80 % of it comes after F_2 burns out. The 2.11×10^{17} cm^{-3} F atoms left then turn Kr_2^* into KrF 17 times faster than Kr_2^* radiates, and each KrF photon gives its F atom back. At Tahoma the row moves the gain from 5.69 to 6.10 %/cm. Kannari's value is no upper bound. It is the low end of a range the report he cites gave, with a condition, for the argon analog. The share saturates, 9.49 % at 1.0×10^{-9} cm^{3}s^{-1}, and at 1.0×10^{-11} it is already 5.85 %. Until the rate is measured the group is a band from 4.65 to 9.49 %, like the trimer law.
- F + F + Kr is held at its 300 K value, which overstates it in hot gas.
- Hakuta's C-A decays and Burnham and Searles' τ_B remain unexplained conflicts.

A Reaction rates

Rates are the values at 300 K. The column *T* gives the law v0_1 applies to them. Electron-impact rows use $\langle\sigma v\rangle$ over the EEDF, and Section A.3 lists their Maxwellian values. Each row's full citation is in `v0_1/network.py`. The row ids are the ones the code and the system map use.

A.1 The basic network, 37 rows

row	reaction	rate	*T*	source
G1	$e^- + Kr \rightarrow e^- + Kr^*$	$\langle\sigma v\rangle$, Table A.3	T_e	Biagi v7.1, the whole 5s manifold at 9.915 eV
C1a	$e^- + Kr \rightarrow e^- + Kr^*$	$\langle\sigma v\rangle$, Table A.3	T_e	Biagi v7.1 5p group at 11.30 eV, prompt cascade to 5s
C1b	$e^- + Kr \rightarrow e^- + Kr^*$	$\langle\sigma v\rangle$, Table A.3	T_e	Biagi v7.1 d+p group at 11.998 eV, as C1a
C1c	$e^- + Kr \rightarrow e^- + Kr^*$	$\langle\sigma v\rangle$, Table A.3	T_e	Biagi v7.1 higher group at 12.75 eV, as C1a
C2	$e^- + Kr \rightarrow 2e^- + Kr^+$	$\langle\sigma v\rangle$, Table A.3	T_e	Biagi v7.1 ionization at 13.996 eV
G6	e^- + Kr, elastic	energy loss $(2m/M)\langle\sigma_m v\varepsilon\rangle$	T_e, T_g	Biagi v7.1 momentum transfer
ATT	$e^- + F_2 \rightarrow F^- + F$	$\langle\sigma v\rangle$, Table A.3	T_e	SIGLO F_2 set
DIS_a3	$e^- + F_2 \rightarrow e^- + 2F$	$\langle\sigma v\rangle$, Table A.3	T_e	SIGLO $a\,^3\Pi_u$, repulsive, so excitation dissociates
DIS_A1	$e^- + F_2 \rightarrow e^- + 2F$	$\langle\sigma v\rangle$, Table A.3	T_e	SIGLO $A\,^1\Pi_u$, as DIS_a3

row	reaction	rate	T	source
VIB1	$e^- + F_2 \rightarrow e^- + F_2$	$\langle \sigma v \rangle$, Table A.3	T_e	SIGLO, v = 1
VIB2	$e^- + F_2 \rightarrow e^- + F_2$	$\langle \sigma v \rangle$, Table A.3	T_e	SIGLO, v = 2
VIB3	$e^- + F_2 \rightarrow e^- + F_2$	$\langle \sigma v \rangle$, Table A.3	T_e	SIGLO, v = 3
VIB4	$e^- + F_2 \rightarrow e^- + F_2$	$\langle \sigma v \rangle$, Table A.3	T_e	SIGLO, v = 4
G2	$2Kr^* \rightarrow Kr^+ + Kr + e^-$	5.0×10^{-10} cm^3s^{-1}		Kannari 1985 Table I, Penning block
G3	$Kr^* + F_2 \rightarrow KrF(B) + F$	4.73×10^{-10} cm^3s^{-1}		rate from Velazco, Kolts and Setser 1976 and Ueda 1984, KrF^* yield 0.90 from Velazco, Kolts and Setser 1976, B share from Yu and Setser 1990
G3C	$Kr^* + F_2 \rightarrow KrF(C) + F$	2.9×10^{-10} cm^3s^{-1}		as G3, with Yu and Setser's k_B/k_C of 1.6
G3X	$Kr^* + F_2 \rightarrow Kr + 2F$	8.47×10^{-11} cm^3s^{-1}		as G3, the KrF^*-free rest, assigned to Kr + 2F
K15	$Kr^+ + 2Kr \rightarrow Kr_2^+ + Kr$	2.4×10^{-31} cm^6s^{-1}	$(T_g/300)^{-0.75}$	the mean of three measurements, 2.4×10^{-31}. T_g law from Perez-Rios and Greene 2018
R11p	$e^- + Kr_2^+ \rightarrow Kr^* + Kr$	1.6×10^{-6} $(T_e/300\,K)^{-0.55}$ cm^3s^{-1}	$T_e^{-0.55}$	Shiu and Biondi 1977, 1.6×10^{-6} $(300\,K/T_e)^{0.55}$, $\pm$ 10 %. Mikus 1978 within 8 %
K26	$Kr^+ + F^- \rightarrow KrF(B)$	2.95×10^{-6} cm^3s^{-1} at 1 atm, 300 K, unscreened	N, screening, $(T_g/300)^{-3}$ on the 3-body part	Kannari 1985, Flannery turnover with Debye screening, and $(T_g/300)^{-3}$ on its three-body part from Thomson theory. Measured with F^- only below 300 Torr, Jacob et al. 1978
K108	$Kr_2^+ + F^- \rightarrow KrF(B) + Kr$	2.95×10^{-6} cm^3s^{-1} at 1 atm, 300 K, unscreened	N, screening, $(T_g/300)^{-3}$ on the 3-body part	as K26
G4	$KrF(B) \rightarrow Kr + F + h\nu$	1.38×10^{8} s^{-1}		Kannari 1985 Table I, the 96.6 % B-X share of τ_B = 6.99 ns. Eden 1978 measured 6.8 $\pm$ 0.2 ns
G4A	$KrF(B) \rightarrow Kr + F + h\nu$	4.84×10^{6} s^{-1}		$A_{B\text{-}A}/A_{B\text{-}X}$ = 0.035, Dunning and Hay's calculation via Yu and Setser 1990. 272 nm is a theoretical position
G4C	$KrF(C) \rightarrow Kr + F + h\nu$	2.0×10^{7} s^{-1}		ML joint fit. Hakuta 1990 measured 60 $\pm$ 10 ns. Band at 275 nm, printed
BC	$KrF(B) + Kr \rightarrow KrF(C) + Kr$	1.05×10^{-11} cm^3s^{-1}		ML joint fit. Hakuta 1990's 1.2×10^{-10} is held out
CB	$KrF(C) + Kr \rightarrow KrF(B) + Kr$	9.55×10^{-12} cm^3s^{-1}	$1/K_{eq}(T_g)$	detailed balance of BC, with the joint fit's K_{eq} and Yu and Setser's measured 120 $\pm$ 50 cm^{-1} gap
G5	$KrF(B) + 2Kr \rightarrow Kr_2F^* + Kr$	1.9×10^{-30} cm^6s^{-1}	$(T_g/300)^{-3.32}$	ML joint fit. T_g law from Shui's calculation for 300-800 K, measured only to 425 K
G5C	$KrF(C) + 2Kr \rightarrow Kr_2F^* + Kr$	3.79×10^{-31} cm^6s^{-1}	$(T_g/300)^{-3.32}$	ML joint fit
QE	$KrF(B) + e^- \rightarrow Kr + F + e^-$	6.6×10^{-8} cm^3s^{-1}		Trainor and Jacob's measured product, re-read with this network's lifetime
QEC	$KrF(C) + e^- \rightarrow Kr + F + e^-$	6.6×10^{-8} cm^3s^{-1}		as QE
QF2	$KrF(B) + F_2 \rightarrow Kr + 3F$	4.75×10^{-10} cm^3s^{-1}		ML joint fit. Xu 1993 printed 3.0 $\pm$ 0.3×10^{-10} with fast mixing
QF2C	$KrF(C) + F_2 \rightarrow Kr + 3F$	3.04×10^{-10} cm^3s^{-1}		ML joint fit
QKR	$KrF(B) + Kr \rightarrow 2Kr + F$	1.0×10^{-14} cm^3s^{-1}		ML joint fit, at its lower bound
QKRC	$KrF(C) + Kr \rightarrow 2Kr + F$	8.37×10^{-13} cm^3s^{-1}		ML joint fit

row	reaction	rate	T	source
C3	$Kr_2F^* \rightarrow 2Kr + F + h\nu$	4.5×10^{6} s^{-1}		Xu, Gadomski and Setser 1993, 220 $\pm$ 20 ns. Band at 420 nm, printed
Q2F	$Kr_2F^* + F_2 \rightarrow 2Kr + F + F_2$	1.2×10^{-10} cm^3s^{-1}		Xu 1993 Table I, $1.2 \pm 0.2\times10^{-10}$
FF	$2F + Kr \rightarrow F_2 + Kr$	2.8×10^{-34} cm^6s^{-1}		Cohen and Bott 1976 evaluation, 2.8×10^{-34}, every rare gas given the Ar value. Held at its 300 K value

A.2 Optional groups on the basic network, 8 rows

The groups kr2, electron are off at the reference point and switch on with `--add`. The first four Kr_2^* rows rest on measurements, and the even split of $Kr_2^* + F_2$ between its two products is Kannari's. $Kr_2^* + F$ and $Kr_2^* + Kr_2^*$ have no measurement among the sources held.

row	reaction	rate	T	source
AK8_Kr	$Kr^* + 2Kr \rightarrow Kr_2^* + Kr$	1.7×10^{-32} cm^6s^{-1}	$(T_g/300)^{-0.64}$	Ueda 1984 Table V, $1.7 \pm 0.2\times10^{-32}$. T_g law from Wieme and Lenaerts 1981
KR2_RAD	$Kr_2^* \rightarrow 2Kr + h\nu$	3.7×10^{6} s^{-1}		Morikawa 1989, 270 $\pm$ 5 ns, and five other measurements
NK7	$Kr_2^* + F_2 \rightarrow KrF(B) + Kr + F$	1.05×10^{-10} cm^3s^{-1}		half of Quigley and Hughes 1978, $2.1 \pm 0.2\times10^{-10}$. The even split is Kannari's
K2F_BR	$Kr_2^* + F_2 \rightarrow Kr_2F^* + F$	1.05×10^{-10} cm^3s^{-1}		the other half of Quigley and Hughes 1978
K2F_F	$Kr_2^* + F \rightarrow KrF(B) + Kr$	3.0×10^{-10} cm^3s^{-1}		Kannari 1985, an estimate. No measurement exists
PEN_KR2	$2Kr_2^* \rightarrow Kr_2^+ + 2Kr + e^-$	5.0×10^{-10} cm^3s^{-1}		Kannari 1985 Table I, from an unpublished 1973 SRI report that gives Ar_2^* the same value. No measurement held
KSTEP	$e^- + Kr^* \rightarrow Kr^+ + 2e^-$	$1.56\times10^{-7}\ T_e^{0.71}\times e^{-4.09/T_e}$ cm^3s^{-1}	T_e	Chen R9 template, an estimate
SUPER	$e^- + Kr^* \rightarrow e^- + Kr$	from G1 by detailed balance	T_e	detailed balance of G1, statistical weight ratio 1/12

A.3 Electron-impact rate coefficients

Maxwellian $\langle\sigma v\rangle$ in cm^3s^{-1} at the T_e of each column. The last column is the factor the two-temperature tail applies at the reference point's mid-pulse state, T_e = 3.25 eV with the tail at 0.18 T_e. The rows below the rule give the energy an attached electron takes from the electron gas.

row	process	T_e (eV)					tail
		0.5	1	1.5	2	3	factor
G1	Kr, 5s manifold, 9.915 eV	1.24×10^{-17}	3.44×10^{-13}	1.11×10^{-11}	6.41×10^{-11}	3.78×10^{-10}	0.044
C1a	Kr, 5p group, 11.3 eV	5.1×10^{-19}	5.93×10^{-14}	2.97×10^{-12}	2.11×10^{-11}	1.48×10^{-10}	7.03×10^{-3}
C1b	Kr, d+p group, 11.998 eV	2.5×10^{-19}	5.84×10^{-14}	3.77×10^{-12}	3.08×10^{-11}	2.53×10^{-10}	2.41×10^{-3}
C1c	Kr, higher group, 12.75 eV	9.15×10^{-21}	7.73×10^{-15}	8.8×10^{-13}	1.02×10^{-11}	1.29×10^{-10}	2.27×10^{-4}
C2	Kr, ionization, 13.996 eV	8.06×10^{-21}	1.57×10^{-14}	2.28×10^{-12}	2.94×10^{-11}	4.14×10^{-10}	7.99×10^{-5}
ATT	F_2, attachment	6.58×10^{-9}	3.51×10^{-9}	2.25×10^{-9}	1.61×10^{-9}	9.75×10^{-10}	1
DIS_a3	F_2, $a\,^3\Pi_u$, 3.16 eV	3.01×10^{-12}	1.13×10^{-10}	3.95×10^{-10}	7.37×10^{-10}	1.35×10^{-9}	1
DIS_A1	F_2, $A\,^1\Pi_u$, 4.34 eV	2.06×10^{-13}	1.93×10^{-11}	8.88×10^{-11}	1.9×10^{-10}	4.02×10^{-10}	1
VIB1	F_2, v = 1, 0.111 eV	1.77×10^{-9}	1.22×10^{-9}	8.72×10^{-10}	6.61×10^{-10}	4.27×10^{-10}	1
VIB2	F_2, v = 2, 0.219 eV	2.01×10^{-9}	1.49×10^{-9}	1.09×10^{-9}	8.37×10^{-10}	5.47×10^{-10}	1
VIB3	F_2, v = 3, 0.324 eV	1.21×10^{-9}	9.75×10^{-10}	7.35×10^{-10}	5.71×10^{-10}	3.78×10^{-10}	1
VIB4	F_2, v = 4, 0.421 eV	3.83×10^{-10}	3.27×10^{-10}	2.51×10^{-10}	1.96×10^{-10}	1.31×10^{-10}	1
ATT	energy an attached e^- takes, eV	0.37	0.46	0.53	0.59	0.72	
	$\frac{3}{2}T_e$, for comparison	0.75	1.50	2.25	3.00	4.50	

A.4 Where v0_1's v0 network differs

The v0 network keeps the notes' six reactions with REBUILD's values, and its optional groups, quench, kr2, ions, electron, bc, f2e, are REBUILD's. These rows differ from v0_1's basic network or exist only in the v0 network. The rows ending in _C are the bc group's copies for KrF(C).

row	reaction	rate	T	source
G3	$Kr^* + F_2 \rightarrow KrF(B) + F$	8.1×10^{-10} cm^3s^{-1}		Kannari 1985 Table I, all to B. The report it credits gives 7.2×10^{-10}
K15	$Kr^+ + 2Kr \rightarrow Kr_2^+ + Kr$	2.5×10^{-31} cm^6s^{-1}	$(T_g/300)^{-0.75}$	Kannari 1985, Boichenko 2000
R11p	$e^- + Kr_2^+ \rightarrow Kr^* + Kr$	2.24×10^{-7} $T_e^{-1/2}$ cm^3s^{-1}	T_e	Kannari 1985 Table I, 1.39×10^{-6} at 300 K on a $T_e^{-1/2}$ template
K26	$Kr^+ + F^- \rightarrow KrF(B)$	2.95×10^{-6} cm^3s^{-1} at 1 atm, 300 K, unscreened	N, screening	Kannari 1985, Flannery turnover with Debye screening, no T_g law
K108	$Kr_2^+ + F^- \rightarrow KrF(B) + Kr$	2.95×10^{-6} cm^3s^{-1} at 1 atm, 300 K, unscreened	N, screening	as K26
G4	$KrF(B) \rightarrow Kr + F + h\nu$	1.43×10^{8} s^{-1}		Kannari 1985 Table I, all of τ_B as B-X
G5	$KrF(B) + 2Kr \rightarrow Kr_2F^* + Kr$	6.0×10^{-31} cm^6s^{-1}	$(T_g/300)^{-3.32}$	Xu, Gadomski and Setser 1993 Table II, $6.0 \pm 2.0\times10^{-31}$
G5_C	$KrF(C) + 2Kr \rightarrow Kr_2F^* + Kr$	6.0×10^{-31} cm^6s^{-1}	$(T_g/300)^{-3.32}$	as G5, applied to KrF(C)
AK8_Kr	$Kr^* + 2Kr \rightarrow Kr_2^* + Kr$	2.7×10^{-32} cm^6s^{-1}		REBUILD's estimate, Kannari's Ar-partner rate given to Kr
KR2_RAD	$Kr_2^* \rightarrow 2Kr + h\nu$	8.0×10^{7} s^{-1}		Kannari 1985 Table I, 8.0×10^{7} per second
NK7	$Kr_2^* + F_2 \rightarrow KrF(B) + Kr + F$	3.0×10^{-10} cm^3s^{-1}		Kannari 1985 Table I
K2F_BR	$Kr_2^* + F_2 \rightarrow Kr_2F^* + F$	3.0×10^{-10} cm^3s^{-1}		Kannari 1985 Table I
K33	$KrF(B) + Kr \rightarrow 2Kr + F$	4.0×10^{-12} cm^3s^{-1}		Xu, Gadomski and Setser 1993, $4.0 \pm 0.5\times10^{-12}$
K33_C	$KrF(C) + Kr \rightarrow 2Kr + F$	4.0×10^{-12} cm^3s^{-1}		as K33, applied to KrF(C)
K32	$KrF(B) + F_2 \rightarrow Kr + 3F$	3.0×10^{-10} cm^3s^{-1}		Xu, Gadomski and Setser 1993, $3.0 \pm 0.3\times10^{-10}$
K32_C	$KrF(C) + F_2 \rightarrow Kr + 3F$	3.0×10^{-10} cm^3s^{-1}		as K32, applied to KrF(C)
K67	$KrF(B) + e^- \rightarrow Kr + F + e^-$	4.5×10^{-8} cm^3s^{-1}		Kushner, Hanson and Schneider 1989
K67_C	$KrF(C) + e^- \rightarrow Kr + F + e^-$	4.5×10^{-8} cm^3s^{-1}		as K67, applied to KrF(C)
K125	$Kr_2F^* + F_2 \rightarrow 2Kr + F + F_2$	1.5×10^{-10} cm^3s^{-1}		Kannari 1985 Table I
K_2Fe	$Kr_2F^* + e^- \rightarrow 2Kr + F + e^-$	1.0×10^{-7} cm^3s^{-1}		Kannari 1985, estimated there
BC_Kr	$KrF(B) + Kr \rightarrow KrF(C) + Kr$	5.0×10^{-10} cm^3s^{-1}		Kannari's B-C mixing with a Kr partner, an estimate

row	reaction	rate	T	source
CB_Kr	KrF(C) + Kr $\rightarrow$ KrF(B) + Kr	3.41×10^{-10} cm^{3}s^{-1}		detailed balance of BC_Kr at 300 K
URAD_C	KrF(C) $\rightarrow$ Kr + F + $h\nu$	1.33×10^{7} s^{-1}		Kannari 1985 Table I, 1.33×10^{7} per second

A.5 Formation energies

Energies above ground-state Kr and F_2 at rest. Every reaction's energy release is the difference of these numbers, so no row can make or lose energy. c_v is the heat capacity per particle in units of k_B.

species	E_{form} (eV)	c_v	what sets it
Kr	0	1.5	ground state
F_2	0	2.5	ground state
F	0.8	1.5	half the F_2 bond, D_0 = 1.60 eV
Kr*	9.915	1.5	Biagi's 5s threshold, the 5s[3/2]2 level
Kr$^+$	13.996	1.5	Biagi's ionization threshold
KrF(B)	5.856	2.5	$D_e(B)$ = 5.54 eV below the ion pair, Tellinghuisen 1976
KrF(C)	5.84112	2.5	120 cm^{-1} below B, Yu and Setser 1990
Kr_2F^*	5.276	3	0.58 eV below KrF* + Kr, calculated by Huestis et al. 1984
Kr_2^*	9.2	2.5	the 147-nm excimer, an estimated well
Kr_2^+	12.846	2.5	IP(Kr) less $D_0(Kr_2^+)$ = 1.15 eV
F^-	-2.6	1.5	E_{form}(F) less the electron affinity, 3.40 eV